Macroeconomics I — MPE Insper

Mock Exam 4 (Simulado 4) — 2026, 3rd term

Solutions and Grading Rubric

Instructions. Three questions, 10 points in total. Suggested time: 2 hours, closed book, non-programmable calculator allowed. Show every step: a number or a formula without the argument that produces it earns little. When an item says *explain*, *justify* or *show graphically*, the explanation and the graph are graded on their own lines: label the axes and every curve, mark the initial and the new point, and indicate the direction of every shift. Where an item lists several approaches or several choice variables, answer *each* of them. Box your final answers. Notation: Kurlat (2020) in Questions 1–2, Benigno (2015) in Question 3.

Question 1 (1.5 points)

A simplified economy has, at first, no government. Values are in R$ million for one year. A coffee farm in Minas Gerais (Firm F) grows beans worth 400 and sells all of them to a roastery (Firm R); F uses no inputs and pays 250 in wages. Firm R buys the 400 of beans and imports packaging worth 100 from abroad; it sells roasted coffee worth 700 to domestic households and worth 200 to foreign buyers, and pays 300 in wages. Each firm keeps whatever is left as profit.

1.a (0.5 point) Calculate the GDP of this economy by the product, the income and the expenditure approaches, explaining precisely the concept used in each. Explain why the imported packaging does not count in GDP, although Firm R used it to produce.

Notation.

- F, R: the coffee farm and the roastery. All values in R$ million for one year.
- $VA_j \geq 0$: value added of firm j, its production minus the intermediate inputs it uses up.
- $W_j \geq 0$: wages paid by firm j. Π_j: its profit (operating surplus), whatever is left after wages; it can be of either sign.
- C, I, G: household consumption, investment and government purchases (final domestic uses). $X \geq 0$, $M \geq 0$: exports and imports. In 1.a, $I = G = 0$.

Step — 0, the definitions used below. GDP is the market value of the *final* goods and services produced in the territory in the year. It is computed three ways, which must agree:

$$\text{GDP} = \textstyle\sum_j VA_j, \tag{G1}$$

$$VA_j = \text{production}_j - \text{intermediate inputs}_j, \tag{G2}$$

$$\Pi_j = VA_j - W_j, \tag{G3}$$

$$\text{GDP} = \textstyle\sum_j (W_j + \Pi_j), \tag{G4}$$

$$\text{GDP} = C + I + G + X - M. \tag{G5}$$

Step — product (value added). F uses no inputs; R uses the beans and the imported packaging. Using (G2) for each firm:

$$VA_F = 400 - 0 = 400, \tag{G6}$$

$$\begin{aligned} VA_R &= (700 + 200) - 400 - 100 \\ &= 900 - 500 \\ &= 400. \end{aligned} \tag{G7}$$

Summing (G6) and (G7) with (G1):

$$\text{GDP} = 400 + 400 = \boxed{800}. \tag{G8}$$

Step — income (factor payments). Profit is what is left of value added after wages. Using (G3) with (G6) and (G7):

$$\Pi_F = 400 - 250 = 150, \tag{G9}$$

$$\Pi_R = 400 - 300 = 100. \tag{G10}$$

Add wages and profits, then apply (G4):

$$\begin{aligned} W_F + W_R &= 250 + 300 = 550, \\ \Pi_F + \Pi_R &= 150 + 100 = 250, \\ \text{GDP} &= 550 + 250 = \boxed{800}. \end{aligned}$$

Step — expenditure (final uses). The final uses are $C = 700$, $I = 0$, $G = 0$, $X = 200$ and $M = 100$. By (G5):

$$\text{GDP} = 700 + 0 + 0 + 200 - 100 = \boxed{800}.$$

Step — imports. The packaging was produced abroad: its 100 is *foreign* value added. In the product approach it is an intermediate input of R and is subtracted from R's production in (G7). In the expenditure approach (G5), C and X are measured at purchaser prices and already contain the packaging, so $-M$ takes it back out. GDP measures what is produced *in the territory*.

Adding sales, $400 + 900 = 1300$, counts the beans twice (once as F's sale, once inside R's price) and counts foreign packaging. The three approaches agree because they are three ways of adding up the same value added.

Intuition. Every real that a final buyer pays for coffee ends up as somebody's income: F's workers and owners, R's workers and owners, or the foreign packaging producer. Take the foreigner out and the three columns must match.

Ref: Kurlat §1.1, pp. 15–21 (value added, the three approaches, imports).

Grading rubric.

- Each approach with the right concept and the number 800: 0.15 pt each (0.45).
- Why imports are excluded (foreign value added; $-M$ corrects purchaser prices): 0.05 pt.
- Only one approach, number right: max. 0.2 pt. Summing sales (1300) or forgetting $-M$ (900): 0 for that line.

1.b (0.5 point) Suppose instead that in the same year households buy only 600 of roasted coffee. Firm R produces exactly as before and keeps the remaining roasted coffee in its warehouse at the end of the year, valued at its selling price. Nothing else changes. Calculate, justifying, the GDP by the product, income and expenditure approaches, compare it with the value in 1.a and explain what the comparison indicates. Then explain what the sale of that stock to households in the following year adds to the following year's GDP, other things equal.

Notation.

- As in 1.a. New: Δinventories, the change in R's stock of unsold roasted coffee during the year, valued at its selling price. It is part of I and can be negative (a drawdown).

Step — what changed. R's production is unchanged; only its destination moved: households, exports, and the part left unsold in the warehouse.

$$\text{production}_R = 600 + 200 + 100 = 900. \tag{G11}$$

An unsold good that was produced is a **change in inventories**, a component of investment:

$$I = \Delta\text{inventories} = 100. \tag{G12}$$

Step — product. F is unchanged, $VA_F = 400$ by (G6). Using (G2) with (G11):

$$\begin{aligned} VA_R &= 900 - 400 - 100 \\ &= 400. \end{aligned}$$

By (G1), GDP $= 400 + 400 = \boxed{800}$.

Step — income. R's operating surplus is measured on production, the stock valued at 100. Using (G3):

$$\begin{aligned}\Pi_R &= 900 - 500 - 300 \\ &= 100.\end{aligned}$$

Wages are $250 + 300 = 550$ and profits $150 + 100 = 250$, with Π_F from (G9). By (G4), GDP $= 550 + 250 = \boxed{800}$.

Step — expenditure. Now $C = 600$, $I = 100$ from (G12), $G = 0$, $X = 200$, $M = 100$. By (G5):

$$\text{GDP} = 600 + 100 + 0 + 200 - 100 = \boxed{800}.$$

Step — comparison. GDP is the same as (G8): **GDP measures production, not sales**. Only the composition of final demand moved, $C \downarrow 100$ and $I \uparrow 100$. A firm that cannot sell is recorded as having invested in its own stock (involuntary, in this case).

Step — next year. Households buy the 100 from stock, so C rises by 100 and inventory investment is -100 (a drawdown). The net contribution to next year's GDP, by (G5):

$$\begin{aligned}\Delta C &= +100, \\ \Delta I = \Delta\text{inventories} &= -100, \\ \Delta\text{GDP} &= 100 - 100 = \boxed{0}.\end{aligned}$$

The coffee was counted in the year it was produced; counting it again when sold would double-count it.

Intuition. Inventories are the entry that makes the expenditure column add up when sales and production differ. Without it the expenditure side would read $600 + 200 - 100 = 700 \neq 800$.

Ref: Kurlat §1.1, p. 18 (inventories and their drawdown), pp. 15–21.

Grading rubric.

- Three approaches with 800 and the inventory entry in I: 0.1 pt each (0.3).
- Comparison: production vs sales, C replaced by I: 0.1 pt.
- Next year: $+C$, $-I$, net zero, with the reason: 0.1 pt.
- GDP = 700 (unsold output treated as not produced): max. 0.1 pt. One approach only: max. 0.25 pt.

1.c (0.5 point) Return to the situation of 1.a. The government now collects a lump-sum tax of 200 from households and uses it to: (i) pay 120 in wages to nurses in public clinics, which treat patients free of charge; (ii) buy 30 of imported medicines, handed out free at those clinics; (iii) pay 50 in cash transfers to low-income families (Bolsa Família). Households' purchases of roasted coffee stay at 700 and nothing else changes. Calculate, justifying, the GDP by the product, income and expenditure approaches, compare it with 1.a and explain what the comparison indicates. Explain why GDP does not rise by 200.

Notation.

- As in 1.a. New: the public clinics as a producer, with value added VA_{clinics}.
- $T = 200$: the lump-sum tax. $TR = 50$: the cash transfers. Both are flows between households and government with nothing produced in exchange.

Step — valuing a free service. Public health care has no market price, so its output is valued **at cost**, wages plus inputs:

$$\text{output}_{\text{clinics}} = 120 + 30 = 150. \tag{G13}$$

The medicines are an intermediate input of the clinics. Using (G2) with (G13):

$$VA_{\text{clinics}} = 150 - 30 = 120. \tag{G14}$$

Step — product. Sum (G6), (G7) and (G14) with (G1):

$$\text{GDP} = 400 + 400 + 120 = \boxed{920}. \tag{G15}$$

Step — income. The clinics have no operating surplus: output is defined as cost, so by (G3) their profit is $120 - 120 = 0$. Adding up and applying (G4):

$$\begin{aligned} \text{wages} &= 250 + 300 + 120 = 670, \\ \text{profits} &= 150 + 100 + 0 = 250, \\ \text{GDP} &= 670 + 250 = \boxed{920}. \end{aligned}$$

Step — expenditure. $G = 150$ is the clinics' output (G13), consumed on behalf of households. Imports now include the medicines:

$$M = 100 + 30 = 130.$$

With $C = 700$, $I = 0$ and $X = 200$, by (G5):

$$\text{GDP} = 700 + 0 + 150 + 200 - 130 = \boxed{920}.$$

Step — what is left out, and why. The tax (200) is a transfer from households to the government, not a payment for a good: it is in no column. **Transfers** (50) buy nothing in exchange: they are not in G and not factor income (if the families spent them, the spending would appear in C; here C is unchanged by assumption). **Imported medicines** (30) enter G but are removed by $-M$: foreign value added.

Step — comparison. Subtract (G8) from (G15):

$$\Delta\text{GDP} = 920 - 800 = \boxed{120},$$

the value added of public health care (G14), i.e. the nurses' wages. Not by 200 (the tax is a financing flow), not by 150 (30 of G is imported), not by 50 (transfers are redistribution).

Intuition. Valuing government output at cost has a consequence: measured public-sector value added equals its wage bill. If the nurses treated twice as many patients for the same pay, GDP would not record it.

Ref: Kurlat §1.1, p. 20 (Example 1.7, government output at cost; Example 1.8, transfers), pp. 15–21.

Grading rubric.

- Three approaches with 920, clinics' output at cost and $VA = 120$: 0.1 pt each (0.3).
- Tax, transfers and imported medicines each correctly excluded, with the reason: 0.1 pt.
- Comparison: +120 is the nurses' value added: 0.1 pt.
- Transfers counted in G (GDP 970) or tax counted (1000): max. 0.2 pt. Medicines not netted by M (950): −0.1.

Gap targeted: Prova 01, 1.b–1.c: the three approaches were asked and only one was given; the classification of a final use (there: software as investment; here: inventories as investment, clinics' output as G).

Question 2 (5.5 points)

Consider an economy described by the Solow model, in which the aggregate variables satisfy:

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}, \tag{1}$$
$$K_{t+1} = (1-\delta)K_t + s\,Y_t, \tag{2}$$
$$L_{t+1} = (1+n)L_t, \tag{3}$$
$$A_{t+1} = (1+g)A_t, \tag{4}$$

with $\{\alpha, \delta, s\} \in (0,1)$ and $n, g \geq 0$.

2.a (1 point) Define the stationary variables, per efficiency unit of labour, with a tilde ($\tilde{k}_t$, $\tilde{y}_t$), and find an expression for $\tilde{k}_{t+1}/\tilde{k}_t$ and then the steady-state value of $\tilde{k}$, denoted $\tilde{k}_{SS}$. Show the steady state graphically. Along the steady state, at what rates do output per worker $y_t = Y_t/L_t$ and aggregate output Y_t grow? Justify, and explain why output per worker is not a stationary variable here.

Notation.

- Y_t, K_t, $C_t > 0$: aggregate output, capital and consumption. $L_t > 0$: number of workers.
- $A_t > 0$: labour-augmenting technology; $A_t L_t$ is labour in efficiency units.
- $\alpha \in (0,1)$: elasticity of output with respect to capital (capital's income share). $\delta \in (0,1)$: depreciation rate. $s \in (0,1)$: saving rate. $n \geq 0$: labour-force growth rate. $g \geq 0$: growth rate of technology.
- $\tilde{k}_t$, $\tilde{y}_t$, $\tilde{c}_t > 0$: capital, output and consumption per efficiency unit of labour. $k_t = K_t/L_t$, $y_t = Y_t/L_t$: per worker.
- $\xi \equiv (1+g)(1+n) - (1-\delta) > 0$: break-even investment per unit of $\tilde{k}$ (defined in step 3). $\tilde{k}_{SS}$, $\tilde{y}_{SS}$: steady-state values.

Step — stationary variables. Technology and labour both grow, so the variables that settle down are those per efficiency unit:

$$\tilde{k}_t \equiv \frac{K_t}{A_t L_t}, \qquad \tilde{y}_t \equiv \frac{Y_t}{A_t L_t}, \qquad \tilde{c}_t \equiv \frac{C_t}{A_t L_t}. \tag{5}$$

Divide (1) by $A_t L_t$ and use (5):

$$\begin{aligned} \tilde{y}_t &= \frac{K_t^{\alpha}(A_t L_t)^{1-\alpha}}{A_t L_t} \\ &= K_t^{\alpha}(A_t L_t)^{-\alpha} \\ &= \left(\frac{K_t}{A_t L_t}\right)^{\alpha} \\ &= \tilde{k}_t^{\alpha}. \end{aligned} \tag{6}$$

Step — law of motion. Multiply (4) by (3):

$$\begin{aligned} A_{t+1}L_{t+1} &= \big[(1+g)A_t\big]\big[(1+n)L_t\big] \\ &= (1+g)(1+n)A_t L_t. \end{aligned} \tag{7}$$

Divide (2) by (7). The second line splits the fraction, the third uses (5), the fourth (6):

$$\begin{aligned} \tilde{k}_{t+1} &= \frac{(1-\delta)K_t + sY_t}{(1+g)(1+n)A_t L_t} \\ &= \frac{(1-\delta)\frac{K_t}{A_t L_t} + s\frac{Y_t}{A_t L_t}}{(1+g)(1+n)} \\ &= \frac{(1-\delta)\tilde{k}_t + s\tilde{y}_t}{(1+g)(1+n)} \end{aligned}$$

$$= \frac{(1-\delta)\tilde{k}_t + s\tilde{k}_t^{\alpha}}{(1+g)(1+n)}. \tag{8}$$

Divide (8) by $\tilde{k}_t$:

$$\boxed{\frac{\tilde{k}_{t+1}}{\tilde{k}_t} = \frac{1-\delta+s\,\tilde{k}_t^{\alpha-1}}{(1+g)(1+n)}} \tag{9}$$

Step — steady state. Set (9) equal to 1 and isolate the term in $\tilde{k}$:

$$\begin{aligned} 1 &= \frac{1-\delta+s\tilde{k}^{\alpha-1}}{(1+g)(1+n)} \\ (1+g)(1+n) &= 1-\delta+s\tilde{k}^{\alpha-1} \\ s\tilde{k}^{\alpha-1} &= (1+g)(1+n)-(1-\delta). \end{aligned}$$

Call the right side ξ; expanding the product,

$$\xi \equiv (1+g)(1+n)-(1-\delta) = n+g+ng+\delta \ (\approx n+g+\delta). \tag{10}$$

Using (10):

$$\begin{aligned} s\tilde{k}^{\alpha-1} &= \xi \\ \tilde{k}^{1-\alpha} &= \frac{s}{\xi}. \end{aligned}$$

Raise both sides to the power $1/(1-\alpha)$:

$$\boxed{\tilde{k}_{SS} = \left(\frac{s}{(1+g)(1+n)-(1-\delta)}\right)^{\frac{1}{1-\alpha}}} \tag{11}$$

Evaluate (6) at (11), writing its denominator as ξ by (10):

$$\begin{aligned} \tilde{y}_{SS} &= \tilde{k}_{SS}^{\alpha} \\ &= \left[\left(\frac{s}{\xi}\right)^{\frac{1}{1-\alpha}}\right]^{\alpha} \\ &= \left(\frac{s}{\xi}\right)^{\frac{\alpha}{1-\alpha}}. \end{aligned} \tag{12}$$

Multiplying the condition $s\tilde{k}^{\alpha-1} = \xi$ by $\tilde{k}$ gives the steady state as investment equal to break-even investment:

$$s\tilde{k}_{SS}^{\alpha} = \xi\,\tilde{k}_{SS}. \tag{13}$$

Step — stability. In (9), $\alpha - 1 < 0$, so $\tilde{k}_{t+1}/\tilde{k}_t$ falls as $\tilde{k}_t$ rises: it is > 1 below $\tilde{k}_{SS}$ and < 1 above. The positive steady state is unique and every $\tilde{k}_0 > 0$ converges to it. Graphically (Figure 1): investment per efficiency unit $s\tilde{k}^{\alpha}$ against break-even investment $\xi\tilde{k}$, the two sides of (13); ξ is the amount needed to equip the new workers (n), the new efficiency units (g) and to replace depreciation (δ).

Step — growth rates. In the steady state $\tilde{y} = \tilde{y}_{SS}$ is constant. Invert (5):

$$y_t = \frac{Y_t}{L_t} = A_t\,\tilde{y}_{SS}, \tag{14}$$

$$Y_t = A_t L_t\,\tilde{y}_{SS}. \tag{15}$$

Take ratios of consecutive periods, using (4) in (14) and (7) in (15):

$$\frac{y_{t+1}}{y_t} = \frac{A_{t+1}\,\tilde{y}_{SS}}{A_t\,\tilde{y}_{SS}}$$

$$= \frac{[(1+g)A_t]\tilde{y}_{SS}}{A_t\,\tilde{y}_{SS}}$$
$$= 1+g \qquad \Rightarrow\ y_t \text{ grows at } \boxed{g},$$
$$\frac{Y_{t+1}}{Y_t} = \frac{A_{t+1}L_{t+1}\,\tilde{y}_{SS}}{A_tL_t\,\tilde{y}_{SS}}$$
$$= \frac{[(1+g)(1+n)A_tL_t]\tilde{y}_{SS}}{A_tL_t\,\tilde{y}_{SS}}$$
$$= (1+g)(1+n) \qquad \Rightarrow\ Y_t \text{ grows at } \boxed{(1+g)(1+n)-1 = g+n+gn}.$$

The same holds for $k_t = K_t/L_t$ (g) and K_t ($g+n+gn$); K/Y is constant.

Step — why y is not stationary. Each period every worker supplies $1+g$ times more efficiency units. Capital per worker must grow at g just to keep pace, so per-worker variables have a trend; dividing by A_tL_t removes it. Long-run growth of y comes only from g, which the model takes as given.

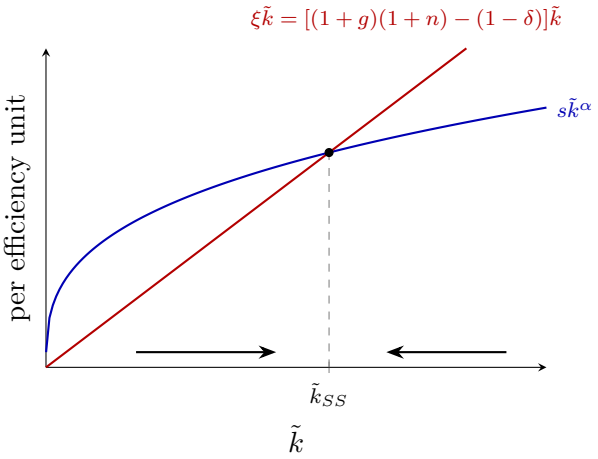


Figure 1. Steady state per efficiency unit. Left of $\tilde{k}_{SS}$ investment exceeds break-even investment and $\tilde{k}$ rises; right of it, $\tilde{k}$ falls. (Drawn for $\alpha = 1/3$, $s = 0.2$, $\xi = 0.1$.)

Intuition. Solow with technological progress is the no-progress model with one change of units: count workers in efficiency units and the economy settles at a constant $\tilde{k}$; in per-worker units that same point is a path growing at g.

Ref: Kurlat §4.2, pp. 57–61 (law of motion, steady state, stability); §4.5, pp. 69–72 (efficiency units, balanced growth path).

Grading rubric.

- Stationary variables defined and $\tilde{y} = \tilde{k}^{\alpha}$: 0.2 pt.
- Ratio $\tilde{k}_{t+1}/\tilde{k}_t$ with $(1+g)(1+n)$ in the denominator: 0.3 pt.
- $\tilde{k}_{SS}$: 0.2 pt. Diagram (axes, both curves named, $\tilde{k}_{SS}$ marked): 0.1 pt.
- Growth rates of y and Y with the reason: 0.2 pt.
- Dividing (2) by A_tL_t instead of $A_{t+1}L_{t+1}$ (no $(1+g)(1+n)$): max. 0.5 pt. Working with K/L as if stationary: max. 0.4 pt.

2.b (1 point) Show which saving rate maximises steady-state consumption per efficiency unit of labour (the golden-rule rate, s_{GR}). Explain why the same s_{GR} also maximises consumption per worker at every date along the steady state, and how a faster rate of technological progress g changes s_{GR} and $\tilde{k}_{GR}$. Give the economic intuition.

Notation.

- As in 2.a. New: $\tilde{c}_{SS}(s)$, steady-state consumption per efficiency unit as a function of the saving rate.
- $f(\tilde{k}) \equiv \tilde{k}^{\alpha}$: output per efficiency unit, by (6).
- $s_{GR} \in (0,1)$, $\tilde{k}_{GR} > 0$: golden-rule saving rate and capital per efficiency unit.
- $c_t \equiv C_t/L_t$: consumption per worker. $A_0 > 0$: technology at date 0.

Step — consumption in steady state. Consumption is the part of output not saved:

$$\tilde{c}_t = (1-s)\,\tilde{y}_t. \tag{16}$$

Substitute (12) into (16), carrying the exponent α of $\tilde{k}_{SS}^{\alpha}$ through the substitution:

$$\begin{aligned}\tilde{c}_{SS}(s) &= (1-s)\,\tilde{k}_{SS}^{\alpha} \\ &= (1-s)\left[\left(\frac{s}{\xi}\right)^{\frac{1}{1-\alpha}}\right]^{\alpha} \\ &= (1-s)\left(\frac{s}{\xi}\right)^{\frac{\alpha}{1-\alpha}}.\end{aligned} \tag{17}$$

Step — first-order condition. Take the log of (17):

$$\ln \tilde{c}_{SS} = \ln(1-s) + \frac{\alpha}{1-\alpha}\ln s - \frac{\alpha}{1-\alpha}\ln \xi. \tag{18}$$

Differentiate (18) with respect to s and set it to zero:

$$\frac{d\ln \tilde{c}_{SS}}{ds} = -\frac{1}{1-s} + \frac{\alpha}{(1-\alpha)s} = 0. \tag{19}$$

Multiply (19) by $(1-\alpha)s(1-s) > 0$:

$$\begin{aligned}-(1-\alpha)s + \alpha(1-s) &= 0 \\ \alpha(1-s) &= (1-\alpha)s \\ \alpha - \alpha s &= s - \alpha s \\ \alpha &= s.\end{aligned}$$

Hence

$$\boxed{s_{GR} = \alpha} \tag{20}$$

SOC: differentiate (19) once more:

$$\frac{d^2\ln \tilde{c}_{SS}}{ds^2} = -\frac{1}{(1-s)^2} - \frac{\alpha}{(1-\alpha)s^2} < 0,$$

so (20) is a maximum.

Step — cross-check by $\tilde{k}$. Substitute (6) into (16), then use the steady-state condition (13), $s\tilde{k}^{\alpha} = \xi\tilde{k}$:

$$\begin{aligned}\tilde{c} &= (1-s)(\tilde{k}^{\alpha}) \\ &= \tilde{k}^{\alpha} - s\tilde{k}^{\alpha} \\ &= \tilde{k}^{\alpha} - (\xi\tilde{k}).\end{aligned} \tag{21}$$

Maximise (21) over $\tilde{k}$:

$$\boxed{f'(\tilde{k}_{GR}) = \alpha \tilde{k}_{GR}^{\alpha-1} = \xi = (1+g)(1+n) - (1-\delta)} \tag{22}$$

The marginal product of capital equals break-even investment per unit of capital. Solve (22) for $\tilde{k}_{GR}$:

$$\tilde{k}_{GR}^{1-\alpha} = \frac{\alpha}{\xi}. \tag{23}$$

Solve (13) for s at $\tilde{k}_{GR}$ and use (23) in the third line:

$$\begin{aligned} s &= \frac{\xi \tilde{k}_{GR}}{\tilde{k}_{GR}^{\alpha}} \\ &= \xi\, \tilde{k}_{GR}^{1-\alpha} \\ &= \xi \cdot \frac{\alpha}{\xi} \\ &= \alpha \ \checkmark \end{aligned}$$

Step — per worker, every date. Invert (5) and use (4):

$$\begin{aligned} c_t = \frac{C_t}{L_t} &= A_t\, \tilde{c}_{SS} \\ &= A_0 (1+g)^t\, \tilde{c}_{SS}. \end{aligned} \tag{24}$$

Take logs of (24):

$$\ln c_t = \ln A_0 + t \ln(1+g) + \ln \tilde{c}_{SS}(s). \tag{25}$$

A_t does not depend on s, so the s that maximises $\tilde{c}_{SS}$ raises the whole path (25) by the same amount at every t: the golden rule maximises consumption per worker at *every* date of the steady state.

Step — effect of g. g does not appear in (20): $s_{GR} = \alpha$ is **unchanged**. Solving (23),

$$\tilde{k}_{GR} = \left(\frac{\alpha}{\xi}\right)^{\frac{1}{1-\alpha}},$$

and by (10), $\partial \xi / \partial g = 1 + n > 0$, so $\boxed{\tilde{k}_{GR} \downarrow}$. Faster progress makes keeping $\tilde{k}$ constant more expensive (steeper break-even line), so the consumption-maximising capital per efficiency unit is lower; but the *share* of output to save does not move, because with Cobb–Douglas the capital income $\alpha \tilde{y}$ always equals the break-even investment $\xi \tilde{k}$ exactly at the golden rule.

Step — knife-edge check. $\alpha \to 0$: capital is useless and $s_{GR} \to 0$ in (20). Dropping the exponent (using $(s/\xi)^{1/(1-\alpha)}$ for $\tilde{y}$ in (17)) gives $s = 1/(2-\alpha) \to 1/2$, which fails this test.

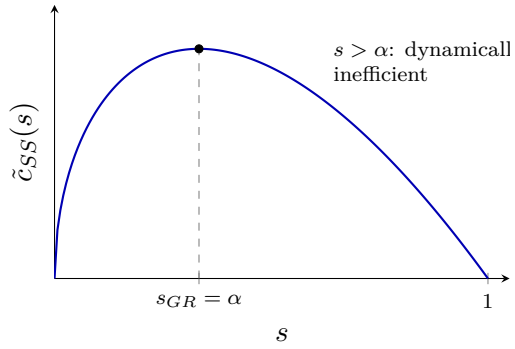


Figure 2. Steady-state consumption per efficiency unit against s (drawn for $\alpha = 1/3$, $\xi = 0.1$): zero at $s = 0$ (nothing to consume from) and at $s = 1$ (nothing consumed), maximum at $s = \alpha$.

Intuition. Saving more raises $\tilde{k}$ and output, but each extra unit of capital also needs ξ of investment every period just to be kept per efficiency unit. Consumption is maximised when the last unit of capital produces exactly what it costs to maintain: $f'(\tilde{k}) = \xi$. Above that point ($s > \alpha$) the economy could consume more today and forever by saving less.

Ref: Kurlat §4.3, pp. 61–63 (golden rule, $s_{gold} = \alpha$); §4.5, pp. 69–72.

Grading rubric.

- $\tilde{c}_{SS}(s)$ with exponent $\alpha/(1-\alpha)$: 0.3 pt.
- FOC and $s_{GR} = \alpha$ (SOC or cross-check $f' = \xi$): 0.3 pt.
- Per-worker consumption at every date maximised: 0.2 pt. Effect of g on s_{GR} and $\tilde{k}_{GR}$: 0.1 pt.
- Intuition (marginal product equals break-even cost): 0.1 pt.
- Exponent dropped in the substitution ($s = 1/(2-\alpha)$): max. 0.4 pt. Maximising $\tilde{k}$ (answer $s = 1$): max. 0.2 pt.

Gap targeted: Prova 01, 2.b: the σ (here α) in the exponent of c_{SS} was dropped.

2.c (1 point) Consider now the competitive equilibrium of an economy that lasts only two periods, $t = 1, 2$. There is an initial capital stock $K_1 > 0$, which belongs to the representative firm and depreciates fully once used in period 1. The representative firm maximises profits in perfectly competitive markets and has production function (1) in both periods, with $A_2 = (1+g)A_1$. The representative consumer owns the firm and receives its profits Π_1 and Π_2. In each period the consumer has one unit of time, which it splits between work L_t and leisure $1 - L_t$, and earns real labour income $w_t A_t L_t$ (w_t is the real wage per efficiency unit of labour). Income not consumed in period 1 is saved as an asset a_1 that pays the real interest rate r; the only asset in this economy is the capital the firm uses in period 2, K_2, which also depreciates fully once used. Intertemporal utility is

$$U = \ln c_1 + \gamma \ln(1 - L_1) + \beta \left[\ln c_2 + \gamma \ln(1 - L_2)\right], \qquad \beta \in (0,1),\ \gamma > 0.$$

Derive and explain the intertemporal budget constraint of this consumer.

Notation.

- $c_1, c_2 > 0$: consumption in periods 1 and 2. $L_t \in [0,1]$: hours worked; $1 - L_t$: leisure (time endowment normalised to 1).
- $A_t > 0$: technology, with $A_2 = (1+g)A_1$. $w_t > 0$: real wage per efficiency unit of labour, so labour income is $w_t A_t L_t$.
- Π_1, Π_2: the firm's profits, paid to the consumer who owns it.
- a_1: asset carried from period 1 to period 2; of either sign for one consumer (in equilibrium it is the capital $K_2 > 0$). $r > -1$: real interest rate between the periods.
- $\beta \in (0,1)$: discount factor. $\gamma > 0$: weight of leisure in utility.
- $H > 0$: full lifetime income, the present value of the time endowment plus profits (step 3).

Step — period constraints. In period 1, uses equal sources:

$$\boxed{c_1 + a_1 = w_1 A_1 L_1 + \Pi_1} \tag{26}$$

Period 2 is the last: nothing is left for later, and nobody would lend to a consumer who cannot repay after period 2. So the consumer spends the asset with its return plus period-2 income:

$$\boxed{c_2 = (1+r)a_1 + w_2 A_2 L_2 + \Pi_2} \tag{27}$$

Profits appear because the consumer owns the firm; with no initial financial wealth, they are its only non-labour income besides the return on a_1.

Step — eliminate a_1. Solve (27) for a_1:

$$a_1 = \frac{c_2 - w_2 A_2 L_2 - \Pi_2}{1+r}. \tag{28}$$

Substitute (28) into (26):

$$c_1 + \frac{c_2 - w_2A_2L_2 - \Pi_2}{1+r} = w_1A_1L_1 + \Pi_1.$$

Move the period-2 income terms to the right side:

$$\boxed{c_1 + \frac{c_2}{1+r} = w_1A_1L_1 + \Pi_1 + \frac{w_2A_2L_2 + \Pi_2}{1+r}} \tag{29}$$

Step — full-income form. Value leisure at its opportunity cost, the wage. Add $w_1A_1(1-L_1) + \frac{w_2A_2(1-L_2)}{1+r}$ to both sides of (29); on the right, $w_tA_tL_t + w_tA_t(1-L_t) = w_tA_t$:

$$\begin{aligned} &c_1 + w_1A_1(1-L_1) + \frac{c_2 + w_2A_2(1-L_2)}{1+r} \\ &\quad = \left[w_1A_1L_1 + w_1A_1(1-L_1)\right] + \Pi_1 + \frac{\left[w_2A_2L_2 + w_2A_2(1-L_2)\right] + \Pi_2}{1+r} \\ &\quad = H, \end{aligned} \tag{30}$$

where

$$H \equiv w_1A_1 + \Pi_1 + \frac{w_2A_2 + \Pi_2}{1+r}. \tag{31}$$

Step — reading. By (30)–(31), the **present value** of lifetime consumption (and of leisure, priced at the wage) equals the present value of lifetime income: the value of the whole time endowment plus profits, with period-2 amounts discounted at $1+r$. a_1 may be of either sign for one consumer; it disappears from the lifetime constraint (29) because borrowing and lending only move resources across periods at the price $1/(1+r)$. The constraint binds: $u' > 0$, so unspent resources at the end are never optimal.

Ref: Kurlat §6.2, pp. 105–112 (intertemporal budget constraint); §7.2, pp. 131–137 (time endowment, labour income); ch. 9, §9.1 (households own the firms).

Grading rubric.

- Two period constraints, with Π_1, Π_2 and the labour income of both periods: 0.4 pt.
- Elimination of a_1 to the present-value constraint: 0.3 pt.
- Explanation as a present-value statement (why $1+r$ discounts, why it binds): 0.3 pt.
- Profits omitted: -0.2. "Consumption adjusted by r" without the present-value reading: max. 0.1 on the last line.

Gap targeted: Prova 01, 2.c: firm profits omitted from the budget constraint; reading not stated in present value.

2.d (1 point) Set up the problem of the consumer of item 2.c and derive the first-order conditions for *all* of its choice variables, including the equation describing the intertemporal consumption decision and the consumption–leisure decision in each period. Then find the equation that gives the optimal c_1 as a function only of parameters, prices and levels of income and wealth, and interpret it economically.

Notation.

- As in 2.c. New: λ_1, $\lambda_2 > 0$, the multipliers on (26) and (27): the marginal utility of one more unit of income in period t.

Step — problem. Choice variables: c_1, c_2, L_1, L_2, a_1; prices w_1, w_2, r and profits Π_1, Π_2 are taken as given. Maximise U subject to (26) and (27), with one multiplier per period constraint:

$$\begin{aligned} \mathcal{L} = {} & \ln c_1 + \gamma \ln(1-L_1) + \beta \ln c_2 + \beta\gamma \ln(1-L_2) \\ & + \lambda_1 \left[w_1A_1L_1 + \Pi_1 - c_1 - a_1\right] + \lambda_2 \left[(1+r)a_1 + w_2A_2L_2 + \Pi_2 - c_2\right]. \end{aligned} \tag{32}$$

Step — FOCs. Differentiate (32) with respect to each choice variable and set the derivative to zero:

$$c_1: \quad \frac{1}{c_1} = \lambda_1, \tag{33}$$

$$c_2: \quad \frac{\beta}{c_2} = \lambda_2, \tag{34}$$

$$L_1: \quad \frac{\gamma}{1-L_1} = \lambda_1 w_1 A_1, \tag{35}$$

$$L_2: \quad \frac{\beta\gamma}{1-L_2} = \lambda_2 w_2 A_2, \tag{36}$$

$$a_1: \quad \lambda_1 = (1+r)\lambda_2. \tag{37}$$

Log utility makes every solution interior: as $L_t \to 1$ the marginal utility of leisure goes to infinity, and as $c_t \to 0$ that of consumption does.

Step — Euler (intertemporal). Substitute (33) and (34) into (37):

$$\left(\frac{1}{c_1}\right) = (1+r)\left(\frac{\beta}{c_2}\right)$$
$$\frac{1}{c_1} = \beta(1+r)\frac{1}{c_2}.$$

Multiply by $c_1 c_2$:

$$\boxed{c_2 = \beta(1+r)\, c_1} \tag{38}$$

The marginal rate of substitution between c_1 and c_2 equals the relative price $1+r$.

Step — consumption–leisure (intratemporal). Divide (35) by (33), and (36) by (34); the multipliers cancel:

$$\frac{\gamma/(1-L_1)}{1/c_1} = \frac{\lambda_1 w_1 A_1}{\lambda_1},$$
$$\frac{\beta\gamma/(1-L_2)}{\beta/c_2} = \frac{\lambda_2 w_2 A_2}{\lambda_2}.$$

$$\boxed{\frac{\gamma c_1}{1-L_1} = w_1 A_1} \tag{39}$$

$$\boxed{\frac{\gamma c_2}{1-L_2} = w_2 A_2} \tag{40}$$

In each period the MRS between leisure and consumption equals the real wage (the price of an hour of leisure).

Step — solve for c_1. Express each term on the left of the full-income constraint (30) in terms of c_1. From (38), (39), and (40) followed by (38):

$$\frac{c_2}{1+r} = \frac{[\beta(1+r)c_1]}{1+r}$$
$$= \beta c_1, \tag{41}$$

$$w_1 A_1 (1-L_1) = \gamma c_1, \tag{42}$$

$$\frac{w_2 A_2 (1-L_2)}{1+r} = \frac{\gamma c_2}{1+r}$$
$$= \frac{\gamma[\beta(1+r)c_1]}{1+r}$$
$$= \beta\gamma c_1. \tag{43}$$

Split the fraction in (30) and substitute (41)–(43):

$$\begin{aligned} c_1 + w_1A_1(1-L_1) + \frac{c_2}{1+r} + \frac{w_2A_2(1-L_2)}{1+r} &= H \\ c_1 + (\gamma c_1) + (\beta c_1) + (\beta\gamma c_1) &= H \\ c_1(1+\gamma+\beta+\beta\gamma) &= H \\ c_1(1+\beta)(1+\gamma) &= H. \end{aligned}$$

Divide by $(1+\beta)(1+\gamma)$ and write H out with (31):

$$\begin{aligned} c_1 &= \frac{H}{(1+\beta)(1+\gamma)} \\ &= \boxed{\frac{1}{(1+\beta)(1+\gamma)}\left[w_1A_1 + \Pi_1 + \frac{w_2A_2+\Pi_2}{1+r}\right]} \end{aligned} \tag{44}$$

Then (38), (39)–(40) and (26) give the other choices:

$$\begin{aligned} c_2 &= \beta(1+r)\,c_1, \\ 1-L_t &= \frac{\gamma c_t}{w_tA_t}, \quad t=1,2, \\ a_1 &= w_1A_1L_1 + \Pi_1 - c_1. \end{aligned}$$

Step — interpretation. By (41)–(44), with log utility the consumer spends fixed shares of full lifetime wealth H: $\frac{1}{(1+\beta)(1+\gamma)}$ on c_1, $\frac{\gamma}{(1+\beta)(1+\gamma)}$ on leisure in period 1, $\frac{\beta}{(1+\beta)(1+\gamma)}$ and $\frac{\beta\gamma}{(1+\beta)(1+\gamma)}$ on period 2. c_1 depends on *lifetime* resources, not on period-1 income alone (permanent income). $\beta\uparrow \Rightarrow c_1\downarrow$ (patience moves spending to period 2); $\gamma\uparrow \Rightarrow c_1\downarrow$ for given H (more of full wealth is taken as leisure). Given H, r does not enter: for the shares, the income and substitution effects of r cancel with log utility; r moves c_1 only through the present value of period-2 income in H (a higher r lowers it, so c_1 falls).

Intuition. Contrast with Prova 01: there work had no utility cost, so $L_1 = 1$ was a corner and saying so was the FOC. Here leisure is valued, so hours are an interior choice and the wage is the price of leisure.

Ref: Kurlat §6.2, pp. 105–115 (Euler equation, log closed forms); §7.2, pp. 131–137 (consumption–leisure); ch. 9, §9.1, eq. (9.1.7)–(9.1.8).

Grading rubric.

- Problem stated with all five choice variables: 0.1 pt.
- FOCs for c_1, c_2 and the Euler equation: 0.3 pt. FOCs for L_1 and L_2 (intratemporal conditions): 0.2 pt (0.1 each).
- FOC for a_1 (or the substitution that eliminates it): 0.1 pt.
- c_1 in closed form: 0.2 pt. Interpretation (H, β, γ, role of r): 0.1 pt.
- No FOC for hours: max. 0.6 pt. Substitution route instead of Lagrangian: full credit.

Gap targeted: Prova 01, 2.d: FOC for L_1 missing ("todas as variáveis de sua escolha").

2.e (1 point) Calculate the saving rate of this representative consumer (the share of saving in the consumer's period-1 income), recalling that markets are perfectly competitive. Find the hours worked in each period. Then find an expression for $\hat{k}_2 \equiv K_2/A_2$ as a function only of $\hat{k}_1 \equiv K_1/A_1$ and of parameters. Treating this expression as a law of motion that applies period after period, draw it with $\hat{k}_1$ on the horizontal axis and $\hat{k}_2$ on the vertical axis, and show graphically the steady state of this economy. Is it stable? Justify.

Notation.

- As in 2.c–2.d. New: Y_1, $Y_2 > 0$, output in each period by (1); $K_2 > 0$, the capital built from period-1 saving.
- $s^* \in (0,1)$: the consumer's saving rate. $\hat{k}_t \equiv K_t/A_t > 0$: capital per unit of technology.
- $\phi(\hat{k})$: the law of motion $\hat{k}_{t+1} = \phi(\hat{k}_t)$. $\hat{k}_{SS}$: its fixed point. $\hat{k}_{GR}$ (Figure 3): the fixed point of the same map with $s = s_{GR}$.

Step — firm, period 1. The firm owns K_1 and hires labour:

$$\max_{L_1} \; K_1^{\alpha}(A_1 L_1)^{1-\alpha} - w_1 A_1 L_1. \tag{45}$$

The FOC of (45), with Y_1 from (1):

$$(1-\alpha)\frac{Y_1}{L_1} = w_1 A_1. \tag{46}$$

Multiply (46) by L_1 to get the wage bill:

$$w_1 A_1 L_1 = (1-\alpha) Y_1. \tag{47}$$

Profit is output minus the wage bill (47):

$$\begin{aligned} \Pi_1 &= Y_1 - w_1 A_1 L_1 \\ &= Y_1 - (1-\alpha) Y_1 \\ &= \boxed{\alpha Y_1}, \end{aligned} \tag{48}$$

the return on the capital the firm owns, paid to its owner.

Step — firm, period 2. The firm borrows K_2 from the consumer and repays $(1+r)K_2$ (the capital is used up):

$$\max_{K_2, L_2} \; K_2^{\alpha}(A_2 L_2)^{1-\alpha} - w_2 A_2 L_2 - (1+r) K_2. \tag{49}$$

The FOCs of (49):

$$\alpha \frac{Y_2}{K_2} = 1 + r, \tag{50}$$

$$(1-\alpha)\frac{Y_2}{L_2} = w_2 A_2. \tag{51}$$

Multiply (50) by K_2 and (51) by L_2 and subtract both payments from output. With constant returns:

$$\begin{aligned} \Pi_2 &= Y_2 - w_2 A_2 L_2 - (1+r) K_2 \\ &= Y_2 - (1-\alpha) Y_2 - \alpha Y_2 \\ &= \boxed{0}. \end{aligned} \tag{52}$$

Step — market clearing. The only asset is the capital of period 2, so the asset market clears when

$$\boxed{a_1 = K_2} \tag{53}$$

The goods markets clear when

$$c_1 + K_2 = Y_1, \tag{54}$$
$$c_2 = Y_2. \tag{55}$$

Labour: L_t supplied $= L_t$ demanded. Check: by (50), (51) and (52), period-2 income in (27) is

$$(1+r)K_2 + w_2 A_2 L_2 + \Pi_2 = \alpha Y_2 + (1-\alpha)Y_2 + 0 = Y_2.$$

Step — saving rate. Substitute (55) and (50) into the Euler equation (38), then multiply by K_2/Y_2:

$$(Y_2) = \beta \left(\frac{\alpha Y_2}{K_2}\right) c_1$$
$$K_2 = \alpha\beta\, c_1. \tag{56}$$

Substitute (56) into the period-1 goods market (54):

$$c_1 + (\alpha\beta c_1) = Y_1$$
$$c_1 = \frac{Y_1}{1+\alpha\beta}. \tag{57}$$

Substitute (57) into (56) and use (53):

$$K_2 = \alpha\beta \left(\frac{Y_1}{1+\alpha\beta}\right)$$
$$a_1 = K_2 = \frac{\alpha\beta}{1+\alpha\beta}\, Y_1. \tag{58}$$

Period-1 income, by (47) and (48):

$$w_1 A_1 L_1 + \Pi_1 = (1-\alpha)Y_1 + \alpha Y_1$$
$$= Y_1. \tag{59}$$

Divide (58) by (59):

$$s^* = \frac{a_1}{w_1 A_1 L_1 + \Pi_1}$$
$$= \frac{\left(\frac{\alpha\beta}{1+\alpha\beta}\, Y_1\right)}{(Y_1)}$$
$$= \boxed{\frac{\alpha\beta}{1+\alpha\beta}} \quad \text{(constant: no } K_1, A_1, g, \gamma\text{)}. \tag{60}$$

Step — hours. Period 2: substitute (55) for c_2 and (51) for $w_2 A_2$ into (40):

$$\frac{\gamma(Y_2)}{1-L_2} = \left(\frac{(1-\alpha)Y_2}{L_2}\right)$$
$$\gamma L_2 = (1-\alpha)(1-L_2)$$
$$L_2(1-\alpha+\gamma) = 1-\alpha$$
$$L_2 = \boxed{\frac{1-\alpha}{1-\alpha+\gamma}}. \tag{61}$$

Period 1: substitute (57) for c_1 and (46) for $w_1 A_1$ into (39):

$$\frac{\gamma\left(\frac{Y_1}{1+\alpha\beta}\right)}{1-L_1} = \left(\frac{(1-\alpha)Y_1}{L_1}\right)$$

$$\begin{aligned}\frac{\gamma Y_1}{(1+\alpha\beta)(1-L_1)} &= \frac{(1-\alpha)Y_1}{L_1}\\ \gamma L_1 &= (1-\alpha)(1+\alpha\beta)(1-L_1)\\ L_1\big[(1-\alpha)(1+\alpha\beta)+\gamma\big] &= (1-\alpha)(1+\alpha\beta)\\ L_1 &= \boxed{\frac{(1-\alpha)(1+\alpha\beta)}{(1-\alpha)(1+\alpha\beta)+\gamma}}. \end{aligned} \tag{62}$$

Neither (61) nor (62) depends on K_1 or A_t: with log utility the income and substitution effects of a higher wage cancel. $L_1 > L_2$: in period 1 consumption is only the share $1/(1+\alpha\beta)$ of output, so a unit of income is worth more at the margin (part of it is saved) and the consumer works more.

Step — law of motion. By (58), (60) and (1):

$$\begin{aligned}K_2 &= s^* Y_1\\ &= s^* K_1^{\alpha}(A_1 L_1)^{1-\alpha}. \end{aligned} \tag{63}$$

Divide (63) by $A_2 = (1+g)A_1$ and use $K_1 = \hat{k}_1 A_1$:

$$\begin{aligned}\hat{k}_2 &= \frac{s^*(\hat{k}_1 A_1)^{\alpha}(A_1 L_1)^{1-\alpha}}{(1+g)A_1}\\ &= \frac{s^* A_1\, \hat{k}_1^{\alpha} L_1^{1-\alpha}}{(1+g)A_1}\\ &= \boxed{\frac{s^* L_1^{1-\alpha}}{1+g}\,\hat{k}_1^{\alpha}} \equiv \phi(\hat{k}_1). \end{aligned} \tag{64}$$

Step — steady state and stability. From (64): $\phi(0) = 0$, $\phi' > 0$, $\phi'' < 0$, $\phi' \to \infty$ as $\hat{k} \to 0$ and $\phi' \to 0$ as $\hat{k} \to \infty$. The curve starts above the 45-degree line and ends below it, so it crosses it exactly once at a positive point. Set $\hat{k} = \phi(\hat{k})$ in (64):

$$\begin{aligned}\hat{k} &= \frac{s^* L_1^{1-\alpha}}{1+g}\,\hat{k}^{\alpha}\\ \hat{k}^{1-\alpha} &= \frac{s^* L_1^{1-\alpha}}{1+g}. \end{aligned}$$

Raise both sides to the power $1/(1-\alpha)$:

$$\boxed{\hat{k}_{SS} = \left(\frac{s^* L_1^{1-\alpha}}{1+g}\right)^{\frac{1}{1-\alpha}}} \tag{65}$$

Differentiate (64):

$$\begin{aligned}\phi'(\hat{k}) &= \alpha\,\frac{s^* L_1^{1-\alpha}}{1+g}\,\hat{k}^{\alpha-1}\\ &= \alpha\,\frac{\phi(\hat{k})}{\hat{k}}. \end{aligned}$$

At the fixed point $\phi(\hat{k}_{SS}) = \hat{k}_{SS}$, so

$$\begin{aligned}\phi'(\hat{k}_{SS}) &= \alpha\,\frac{\phi(\hat{k}_{SS})}{\hat{k}_{SS}}\\ &= \alpha\,\frac{(\hat{k}_{SS})}{\hat{k}_{SS}} \end{aligned}$$

$$= \alpha < 1 \;\Rightarrow\; \text{stable.}$$

From any $\hat{k}_1 < \hat{k}_{SS}$ the staircase between ϕ and the 45-degree line climbs monotonically to $\hat{k}_{SS}$ (Figure 3).

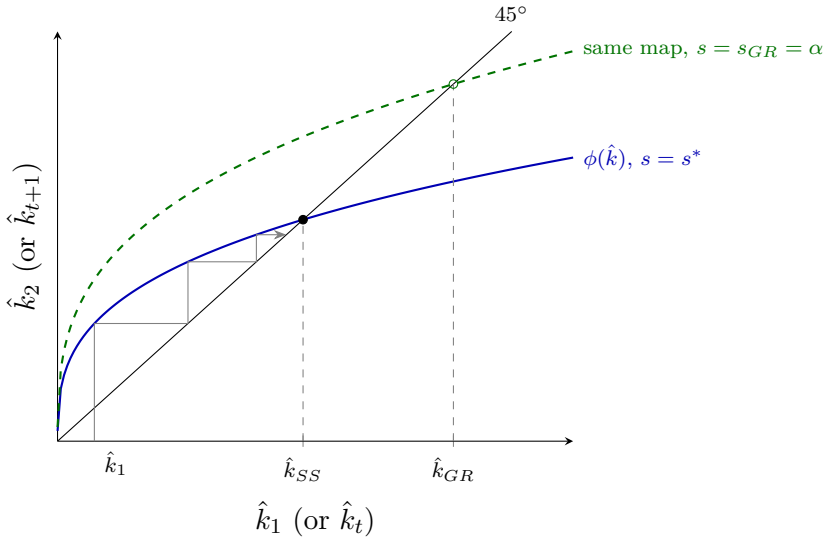


Figure 3. The equilibrium law of motion (solid) crosses the 45-degree line once; the staircase from a low $\hat{k}_1$ converges to $\hat{k}_{SS}$. Dashed: the same map with $s = \alpha$, whose fixed point is the golden rule (used in 2.f). Drawn for $\alpha = 1/3$, $\beta = 0.96$, units chosen so that $\hat{k}_{SS} = 1$.

Intuition. Patience and the productivity of capital set the saving rate; diminishing returns ($\alpha < 1$) make each extra unit of capital produce less, so saving a fixed share of a concave output cannot sustain unbounded growth of $\hat{k}$.

Ref: Kurlat ch. 9, §9.1 (firm FOCs, market clearing, $r = r^K - \delta$ with $\delta = 1$); §4.4, pp. 63–69 (factor shares, zero profit under constant returns); §4.2, pp. 59–61 (stability).

Grading rubric.

- Firm FOCs, $\Pi_1 = \alpha Y_1$, $\Pi_2 = 0$, $1 + r = \alpha Y_2/K_2$: 0.2 pt.
- Market clearing $a_1 = K_2$ (and goods): 0.1 pt. Saving rate $\alpha\beta/(1 + \alpha\beta)$ derived: 0.2 pt.
- Hours L_1, L_2: 0.1 pt. Map $\hat{k}_2(\hat{k}_1)$: 0.2 pt.
- Diagram: axes, 45-degree line, concave map from the origin, $\hat{k}_{SS}$ at the crossing, stability argued: 0.2 pt.
- Saving rate quoted from Solow's s or left as a function of prices: 0 for that line. No market clearing ($K_2 \neq a_1$): max. 0.5 pt. A map that does not cross the 45-degree line: 0 for the diagram.

Gap targeted: Prova 01, 2.e: saving rate not derived; no market clearing $K_2 = a_1$; diagram without a valid steady state.

2.f (0.5 point) Discuss intuitively the relation between the model of items 2.c–2.e and the Solow model of items 2.a–2.b. In particular, compare the saving rate found in 2.e with s_{GR} and explain the difference.

Notation.

- As in 2.a–2.e. New: $\tilde{k}_t \equiv \hat{k}_t/L_1 = K_t/(A_t L_1) > 0$, capital per efficiency unit of *hours* (in 2.a, per efficiency unit of workers).

Step — same law of motion. Write (64), applied period after period, in terms of $\tilde{k}_t = \hat{k}_t/L_1$:

$$L_1 \tilde{k}_{t+1} = \frac{s^* L_1^{1-\alpha}}{1+g} (L_1 \tilde{k}_t)^\alpha$$

$$= \frac{s^* L_1}{1+g}\,\tilde{k}_t^{\alpha}$$
$$(1+g)\tilde{k}_{t+1} = s^*\tilde{k}_t^{\alpha}. \tag{66}$$

Set $\boxed{\delta = 1}$ (capital used up in one period) and $\boxed{n = 0}$ (one consumer) in the Solow law of motion (8), which is equation (2) divided as in 2.a:

$$\tilde{k}_{t+1} = \frac{(1-1)\tilde{k}_t + s\tilde{k}_t^{\alpha}}{(1+g)(1+0)}$$
$$= \frac{s\tilde{k}_t^{\alpha}}{1+g}$$
$$(1+g)\tilde{k}_{t+1} = s\,\tilde{k}_t^{\alpha}. \tag{67}$$

(66) is (67) with $s = s^*$. Indeed, by (65),

$$\hat{k}_{SS} = \left(\frac{s^* L_1^{1-\alpha}}{1+g}\right)^{\frac{1}{1-\alpha}}$$
$$= L_1\left(\frac{s^*}{1+g}\right)^{\frac{1}{1-\alpha}}$$
$$= L_1\left(\frac{s^*}{(1+g)(1+0)-(1-1)}\right)^{\frac{1}{1-\alpha}},$$

which is 2.a's $\tilde{k}_{SS}$ (11) times the hours worked.

Step — what is new. Solow *assumes* s and the labour input. Here both are *chosen*: $s^* = \alpha\beta/(1+\alpha\beta)$ in (60) comes from patience β and the return to capital α, and log utility makes it constant (independent of K, A, g) — a microfoundation for Solow's constant saving rate. Hours L_1 in (62) depend on the taste for leisure γ and on β.

Step — against the golden rule. Compare (60) with $s_{GR} = \alpha$ from (20):

$$s^* < \alpha \iff \frac{\alpha\beta}{1+\alpha\beta} < \alpha$$
$$\iff \frac{\beta}{1+\alpha\beta} < 1$$
$$\iff \beta < 1+\alpha\beta$$
$$\iff \beta(1-\alpha) < 1,$$

always true, since $\beta < 1$ and $1-\alpha < 1$. So $\boxed{s^* < \alpha}$ and $\hat{k}_{SS} < \hat{k}_{GR}$ (Figure 3): the competitive economy accumulates *less* than the golden rule. Even as $\beta \to 1$, $s^* \to \alpha/(1+\alpha) < \alpha$: here saving finances one period of consumption only, since the horizon ends.

Step — why this is not a failure. The golden rule maximises steady-state consumption and ignores that consumption earlier is worth more to an impatient consumer ($\beta < 1$) and that reaching more capital costs consumption on the way. Markets here are complete and competitive, with no externalities: by the first welfare theorem the equilibrium is Pareto efficient, and it is below the golden rule, i.e. dynamically efficient. The golden rule is a property of steady states, not a welfare criterion.

Ref: Kurlat §4.3, pp. 61–63; ch. 9, §9.2 (first welfare theorem) and §9.3 (comparison with the golden rule).

Grading rubric.

- Same law of motion with $\delta = 1$, $n = 0$: 0.1 pt.
- Saving (and hours) chosen, not assumed, and from which primitives: 0.2 pt.
- $s^* < \alpha$ shown, with the reason (impatience; golden rule not a welfare criterion): 0.2 pt.
- "It is Solow with $\delta = 1$ and $n = 0$" and nothing else: max. 0.1 pt.

Gap targeted: Prova 01, 2.f: the endogenous saving rate and its comparison with s_{GR} were missing.

Question 3 (3 points)

Consider the simplified New Keynesian model of Benigno (2015), based on the aggregate demand and supply equations below (barred variables are long-run values, when all firms reset their prices; the others are short-run values):

$$y = \bar{y} + (g - \bar{g}) - \sigma\,[i - (\bar{p} - p)] - \sigma \ln \beta, \tag{AD}$$

$$p - p^e = \kappa\,(y - y_n), \tag{AS}$$

where p^e is the price set by the firms that cannot reset prices and y_n is short-run natural output,

$$y_n = \frac{1+\eta}{\sigma^{-1}+\eta}\,a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}\,g - \frac{1}{\sigma^{-1}+\eta}\,\mu, \tag{YN}$$

μ being the mark-up; $\bar{y}$ is given by the same expression evaluated at the long-run values $\bar{a}$, $\bar{g}$, $\bar{\mu}$; and $y_e = \frac{1+\eta}{\sigma^{-1}+\eta}a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}g$ is the efficient level of output. All variables are log deviations from an efficient steady state. Initially $a = \bar{a} = g = \bar{g} = \mu = \bar{\mu} = 0$, $\bar{p} = p^e$, and the central bank sets $i = \rho \equiv -\ln\beta$, so that $y = y_n = y_e = \bar{y}$ and $p = p^e$.

3.a (1 point) News arrives that productivity will be permanently higher in the long run: $\bar{a}$ rises by $d\bar{a} > 0$, while current productivity a does not change. With i unchanged, represent graphically, explaining economically why, the initial equilibrium and the shift that defines the new equilibrium. Compute dy and dp. Indicate the gap $y - y_n$ and the gap $y - y_e$ and use both in your explanation. Is this boom good for welfare today? Justify.

Notation.

- y, p: output and price level today; $\bar{y}$, $\bar{p}$: their long-run values, when all firms have reset prices. All are log deviations from the efficient steady state.
- p^e: the price preset by the firms that cannot reset this period.
- i: nominal interest rate, the central bank's instrument. $\beta \in (0,1)$: discount factor; $\rho \equiv -\ln\beta > 0$: the initial (neutral) rate.
- a, $\bar{a}$: productivity today and in the long run. g, $\bar{g}$: government spending. μ, $\bar{\mu}$: the firms' mark-up. All start at 0.
- $\sigma > 0$: intertemporal elasticity of substitution. $\eta > 0$: inverse Frisch elasticity of labour supply. $\kappa > 0$: slope of AS.
- y_n: natural output (flexible prices, with the mark-up). y_e: efficient output (no mark-up).
- $\Gamma \equiv \frac{1+\eta}{\sigma^{-1}+\eta} > 0$: response of y_n, y_e and $\bar{y}$ to productivity.

Step — 0, the remaining statement equations. With Γ as defined above, the statement gives long-run output (the expression (YN) at long-run values) and efficient output:

$$\bar{y} = \Gamma\,\bar{a} + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}\,\bar{g} - \frac{1}{\sigma^{-1}+\eta}\,\bar{\mu}, \tag{68}$$

$$y_e = \Gamma\,a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}\,g. \tag{69}$$

Substitute $\rho = -\ln\beta$ into (AD), then use the initial values $g = \bar{g} = 0$ and $\bar{p} = p^e$:

$$\begin{aligned} y &= \bar{y} + (g - \bar{g}) - \sigma\,[i - (\bar{p} - p)] - \sigma(-\rho) \\ &= \bar{y} + (g - \bar{g}) - \sigma(i - \rho) - \sigma(p - \bar{p}) \\ &= \bar{y} + (0 - 0) - \sigma(i - \rho) - \sigma(p - (p^e)) \\ &= \bar{y} - \sigma(i - \rho) - \sigma(p - p^e). \end{aligned} \tag{70}$$

Step — which bar moves which curve. Differentiate (68) with only $\bar{a}$ moving:

$$d\bar{y} = \Gamma\,d\bar{a} > 0. \tag{71}$$

$\bar{y}$ is in AD (70), so AD shifts right by $\Gamma d\bar{a}$ at every p (up by $\Gamma d\bar{a}/\sigma$). y_n in (YN) and y_e in (69) depend on the *current* a, g, μ, none of which moved: AD does not move (it still passes through (y_n, p^e)), and $y_e = y_n = 0$ still.

Step — why AD moves (Euler). Long-run consumption $\bar{c} = \bar{y}$ rises: households expect higher permanent income and, to smooth consumption, want to consume more *today* at the same real rate $i - (\bar{p} - p)$. Current demand for goods rises at every price level.

Step — new equilibrium. With $i = \rho$, (70) and (71) give AD; with $y_n = 0$, (AS) gives AS:

$$\begin{aligned} y &= (\Gamma\, d\bar{a}) - \sigma(\rho - \rho) - \sigma(p - p^e) \\ &= \Gamma\, d\bar{a} - \sigma(p - p^e), \qquad (72) \\ p - p^e &= \kappa\,(y - 0) \\ &= \kappa\, y. \qquad (73) \end{aligned}$$

Substitute (73) into (72):

$$\begin{aligned} y &= \Gamma\, d\bar{a} - \sigma(\kappa\, y) \\ y(1 + \sigma\kappa) &= \Gamma\, d\bar{a}. \end{aligned}$$

Divide by $1 + \sigma\kappa$ (the initial y is 0, so $y = dy$):

$$\boxed{dy = \frac{\Gamma\, d\bar{a}}{1 + \sigma\kappa} > 0} \qquad (74)$$

Substitute (74) into (73):

$$\begin{aligned} dp &= \kappa\, dy \\ &= \kappa \left(\frac{\Gamma\, d\bar{a}}{1 + \sigma\kappa} \right) \\ &= \boxed{\frac{\kappa\, \Gamma\, d\bar{a}}{1 + \sigma\kappa} > 0}. \qquad (75) \end{aligned}$$

Step — the two gaps. Since $y_n = y_e = 0$, both gaps equal (74). $y - y_n = dy > 0$: output above natural, which is why prices rise in (75) (AS: flexible-price firms face higher real marginal cost, because households must be paid a higher real wage to work more). $y - y_e = dy > 0$: output above the efficient level as well. Chain: $\bar{a} \uparrow \rightarrow \bar{y} \uparrow \rightarrow c \uparrow$ (Euler) $\rightarrow$ AD right $\rightarrow y > y_n \rightarrow$ marginal cost $\uparrow \rightarrow$ flexible firms raise $p \rightarrow$ real rate $i - (\bar{p} - p) \uparrow$, which limits the rise in y (movement along AD). Firms with preset prices keep p^e and produce whatever is demanded.

Step — welfare. No. Today's technology is unchanged, so the efficient output today is unchanged. The extra output is produced by working beyond the point where the MRS between leisure and consumption equals today's productivity, and the price surprise $p > p^e$ creates price dispersion across identical firms. Both terms of the welfare loss are positive: the economy is overheating on a forecast.

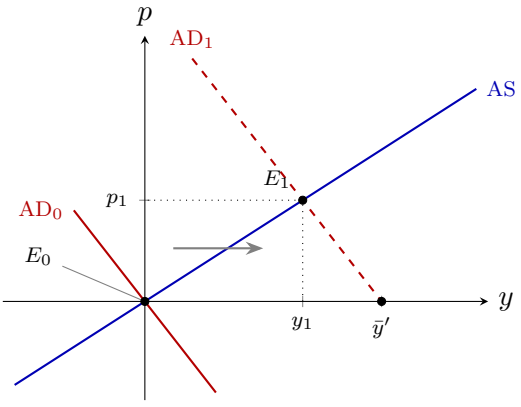


Figure 4. News of higher $\bar{a}$ shifts AD right (it now passes through $(\bar{y}', \bar{p})$ at $i = \rho$); AS stays. E_1: $y_1 - y_n = y_1 - y_e > 0$ and $p_1 > p^e$. (Drawn for $\sigma = 0.5$, $\kappa = 1$, $\Gamma d\bar{a} = 1$; the origin is (y_n, p^e).)

Intuition. Expectations about the future move demand today. Supply today has not changed, so the extra demand can only be met by working harder with the same technology, and prices rise.

Ref: Benigno (2015) §6.3, Fig. 8, pp. 511–513; §5, eqs. (20)–(21), pp. 509–511; §4.2–§4.4 (y_n, y_e), pp. 508–509.

Grading rubric.

- AD shifts right, with the Euler/permanent-income reason: 0.3 pt.
- AS does not move, because y_n depends on current a, μ only: 0.2 pt.
- dy and dp: 0.2 pt.
- Both gaps signed and used in the explanation: 0.2 pt. Welfare verdict with the reason: 0.1 pt.
- Graph lines are inside the first two items (axes y, p; AD_0, AD_1, AS named; E_0, E_1; direction of the shift).
- Shifting AS (treating the news as current productivity): max. 0.3 pt. Signs of dy, dp inconsistent with the graph: -0.2.

3.b (1 point) Based on the previous item, explain intuitively what a monetary authority committed to price stability should do after the news, by how much it must move its instrument, and what is the economic cost of doing so. Show graphically the paths of y and p in the short and in the long run, with and without that response. A commentator claims that "whatever the central bank does, it must give up either price stability or a closed output gap". Evaluate the claim.

Notation.

- As in 3.a. New: r_n, the natural real interest rate, the real rate that clears today's goods market at natural output.

Step — target. Price stability: $p = p^e$. On AS (AS) this requires $y = y_n$, and since $y_n = y_e$ here, it also gives $y = y_e$.

Step — instrument. AD must pass through $(y_n, p^e) = (0, p^e)$. Set $y = 0$ and $p = p^e$ in (70), with $\bar{y} = \Gamma d\bar{a}$ from (71):

$$\begin{aligned} 0 &= (\Gamma\, d\bar{a}) - \sigma(i - \rho) - \sigma(p^e - p^e) \\ 0 &= \Gamma\, d\bar{a} - \sigma(i - \rho) \\ \sigma(i - \rho) &= \Gamma\, d\bar{a}. \end{aligned}$$

Divide by σ:

$$\boxed{i = \rho + \frac{\Gamma\, d\bar{a}}{\sigma}} \tag{76}$$

Raise i by $\Gamma d\bar{a}/\sigma > 0$. The natural real rate is

$$r_n = \rho + \sigma^{-1}(\bar{y} - y_n). \tag{77}$$

Differentiate (77) and substitute (71) and $dy_n = 0$:

$$\begin{aligned} dr_n &= \sigma^{-1}\left[(\Gamma\, d\bar{a}) - (0)\right] \\ &= \frac{\Gamma\, d\bar{a}}{\sigma}. \end{aligned}$$

The rise in (76) is exactly the rise in r_n. The future is richer, so the real rate that clears today's goods market at natural output is higher. AD returns to AD_0 in Figure 4.

Step — cost. None in terms of the two objectives: after the rise $y = y_n = y_e$ and $p = p^e$. The shock is *efficient*: it does not separate y_n from y_e (no mark-up moved), so one instrument closes both gaps (divine coincidence). The commentator is wrong for this shock. A cut, the instinct to "support the good news", would push output and prices further up.

Step — long run. In period 2 all prices adjust and $y = \bar{y}' = \Gamma d\bar{a}$, $p = \bar{p}$, with or without the response. The higher rate does not cancel the gain; it *times* it: households wait for the productivity to arrive instead of working beyond the efficient level today. The same response is right if the news is only optimism; the model cannot tell the two apart in the short run.

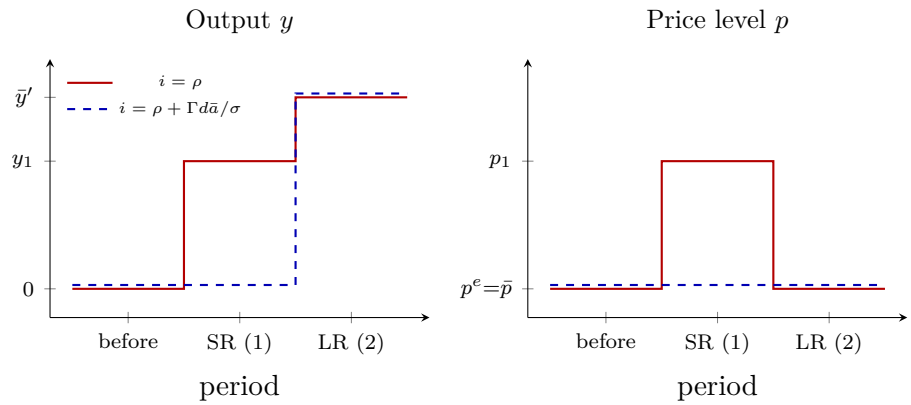


Figure 5. Solid: no response, a boom with a price surprise in the short run. Dashed: the rate rise keeps $y = y_n = y_e$ and $p = p^e$ in the short run; in the long run both paths reach $\bar{y}'$ and $\bar{p}$.

Intuition. The central bank's job is to set the real rate to the one the economy would have with flexible prices. Good news about the future raises that rate; tracking it is not "fighting" the news.

Ref: Benigno (2015) §6.3, Fig. 8, pp. 511–513 (optimal response to expected productivity); §5 (r_n); §11, p. 522 (no trade-off when $y_n = y_e$).

Grading rubric.

- Raise i, with the mechanism (AD back through (y_n, p^e)): 0.3 pt.
- Size $\Gamma d\bar{a}/\sigma$, read as the rise in r_n: 0.2 pt.
- Cost: none, both gaps closed, because the shock is efficient ($y_n = y_e$); claim rejected: 0.3 pt.
- Paths SR/LR with and without the response: 0.2 pt (0.1 if the long run is missing).
- Cutting i: max. 0.2 pt on the item. "Cost: a recession" (a trade-off asserted): max. 0.5 pt.

Gap targeted: Prova 01, 3.a–3.b: which bar moves which curve; sign of the policy response to a pure AD shift; the cost of a response when there is no trade-off.

3.c (1 point) Suppose the monetary authority represents society's preferences by the quadratic loss function, with parameter $\phi \geq 0$,

$$L = \frac{1}{2}(y - y_e)^2 + \frac{\phi}{2\kappa}(p - p^e)^2 . \tag{LF}$$

In the same period as the news of item 3.a, the government temporarily raises a tax on firms' sales revenue (an IPI-like tax), which raises the current mark-up by $d\mu > 0$; the long-run mark-up $\bar{\mu}$ does not change. Derive the optimal monetary-policy rule (IT), find the optimal y, p and i, and show graphically the optimal policy. Discuss how your answer depends on ϕ, separating the part of the optimal response that answers the news from the part that answers the mark-up shock.

Notation.

- As in 3.a–3.b. New: $\phi \geq 0$, the weight on price surprises in (LF).
- $m \equiv \frac{d\mu}{\sigma^{-1}+\eta} > 0$: the fall in natural output caused by the tax. y_n': natural output after the tax.
- y^*, p^*, i^*: optimal output, price level and interest rate; $O = (y^*, p^*)$.
- N, y_N: the point on the new AS, and its output, reached when i only offsets the news.

Step — what the tax moves. Differentiate (YN) with respect to the current mark-up:

$$y_n' = -\frac{d\mu}{\sigma^{-1} + \eta} = -m < 0. \tag{78}$$

Current natural output falls, so **AS shifts up/left** through (y_n', p^e). y_e in (69) has no μ: $y_e = 0$ is unchanged. Now $y_n \neq y_e$: the tax is an *inefficient* shock. $\bar{\mu}$ is unchanged, so by (68) the AD shift is only the news ($\bar{y} = \Gamma d\bar{a}$, (71)).

Step — AS is the constraint. The central bank moves AD with i; for any point on AS there is an i that puts AD through it. AS, with p^e preset, is what policy cannot move within the period: it is the feasible set. So choose y on AS and recover i from AD. By (AS) with (78), the constraint is

$$p - p^e = \kappa\,(y - y_n'). \tag{79}$$

Substitute (79) into (LF):

$$\begin{aligned} L(y) &= \tfrac{1}{2}(y - y_e)^2 + \tfrac{\phi}{2\kappa}\left[\kappa\,(y - y_n')\right]^2 \\ &= \tfrac{1}{2}(y - y_e)^2 + \tfrac{\phi}{2\kappa}\kappa^2(y - y_n')^2 \\ &= \tfrac{1}{2}(y - y_e)^2 + \tfrac{\phi\kappa}{2}(y - y_n')^2. \end{aligned} \tag{80}$$

Step — targeting rule. Differentiate (80) and set it to zero:

$$\frac{dL}{dy} = (y - y_e) + \phi\kappa\,(y - y_n') = 0. \tag{81}$$

By (79), $\kappa(y - y_n') = p - p^e$. Substitute this into (81):

$$\boxed{(y - y_e) + \phi\,(p - p^e) = 0} \qquad \text{(IT: a line through } (y_e, p^e) \text{ with slope } -1/\phi). \tag{82}$$

SOC: from (80), $d^2L/dy^2 = 1 + \phi\kappa > 0$.

Step — optimal point. Solve (81) for y:

$$\begin{aligned} y(1 + \phi\kappa) &= y_e + \phi\kappa\, y_n' \\ y^* &= \frac{y_e + \phi\kappa\, y_n'}{1 + \phi\kappa}. \end{aligned} \tag{83}$$

Substitute $y_e = 0$ and (78) into (83):

$$\begin{aligned} y^* &= \frac{(0) + \phi\kappa\,(-m)}{1 + \phi\kappa} \\ &= \boxed{-\frac{\phi\kappa}{1 + \phi\kappa}\,m < 0} \end{aligned} \tag{84}$$

Substitute (84) and (78) into (79):

$$\begin{aligned} p^* - p^e &= \kappa(y^* - y_n') \\ &= \kappa\left[\left(-\frac{\phi\kappa}{1 + \phi\kappa}\,m\right) - (-m)\right] \\ &= \kappa\left(m - \frac{\phi\kappa\, m}{1 + \phi\kappa}\right) \\ &= \boxed{\frac{\kappa}{1 + \phi\kappa}\,m > 0}. \end{aligned} \tag{85}$$

Check in (82):

$$\begin{aligned} (y^* - y_e) + \phi(p^* - p^e) &= \left[\left(-\frac{\phi\kappa}{1 + \phi\kappa}\,m\right) - (0)\right] + \phi\left(\frac{\kappa}{1 + \phi\kappa}\,m\right) \\ &= \frac{-\phi\kappa m + \phi\kappa m}{1 + \phi\kappa} = 0 \ \checkmark \end{aligned}$$

The bank splits the shock: output below efficient *and* prices above p^e.

Step — optimal rate. Evaluate AD (70) at (y^*, p^*), with $\bar{y} = \Gamma d\bar{a}$, and solve for i^*:

$$\begin{aligned}(y^*) &= (\Gamma\, d\bar{a}) - \sigma(i^* - \rho) - \sigma\big((p^*) - p^e\big)\\ \sigma(i^* - \rho) &= \Gamma\, d\bar{a} - y^* - \sigma(p^* - p^e)\\ i^* &= \rho + \frac{\Gamma\, d\bar{a} - y^*}{\sigma} - (p^* - p^e).\end{aligned} \tag{86}$$

Substitute (84) and (85) into (86):

$$\begin{aligned}i^* &= \rho + \frac{\Gamma\, d\bar{a} - \left(-\frac{\phi\kappa}{1+\phi\kappa}\, m\right)}{\sigma} - \left(\frac{\kappa}{1+\phi\kappa}\, m\right)\\ &= \rho + \frac{\Gamma\, d\bar{a}}{\sigma} + \frac{\phi\kappa m}{\sigma(1+\phi\kappa)} - \frac{\kappa m}{1+\phi\kappa}.\end{aligned} \tag{87}$$

Put the last two terms of (87) over the common denominator $\sigma(1+\phi\kappa)$:

$$\boxed{i^* = \rho + \underbrace{\frac{\Gamma\, d\bar{a}}{\sigma}}_{\text{news}} + \underbrace{\frac{\kappa\, m\, (\phi - \sigma)}{\sigma(1+\phi\kappa)}}_{\text{mark-up}}} \tag{88}$$

Step — the news part does not depend on ϕ. The news moves neither y_n nor y_e, so by (83) it does not move the optimal point $O = (y^*, p^*)$ on AS. The bank simply undoes the AD shift, $di = \Gamma d\bar{a}/\sigma$ in (88), for *any* ϕ, exactly as in (76).

Step — the mark-up part depends on ϕ against σ. Substitute the news-offset rate (76) and $\bar{y} = \Gamma d\bar{a}$ into AD (70): AD is back at AD_0. Then substitute (79) for $p - p^e$ to find the point N:

$$\begin{aligned}y &= (\Gamma\, d\bar{a}) - \sigma\left(\frac{\Gamma\, d\bar{a}}{\sigma}\right) - \sigma(p - p^e)\\ &= -\sigma(p - p^e)\\ y &= -\sigma\left[\kappa\,(y - y_n')\right]\\ y_N &= \frac{\sigma\kappa}{1+\sigma\kappa}\, y_n'.\end{aligned} \tag{89}$$

Subtract (89), with $y_n' = -m$, from (84):

$$\begin{aligned}y^* - y_N &= \left(-\frac{\phi\kappa}{1+\phi\kappa}\, m\right) - \frac{\sigma\kappa}{1+\sigma\kappa}\,(-m)\\ &= -\frac{\phi\kappa m}{1+\phi\kappa} + \frac{\sigma\kappa m}{1+\sigma\kappa}\\ &= \frac{\kappa m(\sigma - \phi)}{(1+\phi\kappa)(1+\sigma\kappa)}.\end{aligned} \tag{90}$$

If $\phi > \sigma$ (the IT line is flatter than AD), (90) is negative: the bank raises i further and accepts a deeper output loss for less price movement; if $\phi < \sigma$ it raises i by less than the news alone requires. Limits of (84), (85) and (88):

$$\begin{aligned}\phi \to \infty \text{ (strict price stability)}: &\quad y^* = y_n', \quad p^* = p^e, \quad i^* = \rho + \frac{\Gamma d\bar{a} + m}{\sigma},\\ \phi = 0: &\quad y^* = y_e, \quad p^* = p^e + \kappa m, \quad i^* = \rho + \frac{\Gamma d\bar{a}}{\sigma} - \kappa m.\end{aligned}$$

Step — trade-off. Unlike 3.b there is a trade-off: $y_n' \neq y_e$, so no rate closes both gaps; ϕ chooses where on AS to sit.

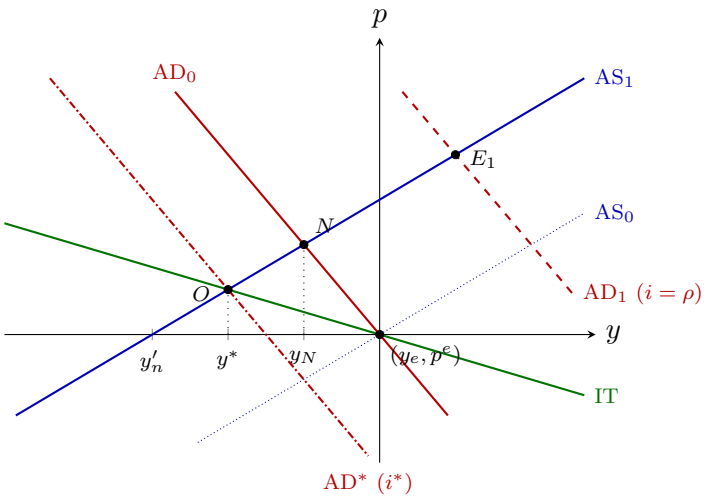


Figure 6. The tax shifts AS from AS_0 to AS_1 (through (y_n', p^e)); the news shifts AD from AD_0 to AD_1. With $i = \rho$ the economy is at E_1. Offsetting the news alone brings AD back to AD_0 and the economy to N. The optimum O is where AS_1 meets the IT line; drawn for $\phi = 2 > \sigma = 0.5$, so the bank raises i beyond the news offset, to AD^*. ($\kappa = 1$, $\Gamma d\bar{a} = m = 1$.)

Intuition. Two shocks, two different jobs. The news is a demand shock with no trade-off: offset it fully. The tax is a cost shock that drives a wedge between natural and efficient output: no rate fixes both gaps, so the convex loss spreads the miss, and ϕ says how much of it goes into prices.

Ref: Benigno (2015) §11, eqs. (33)–(35), Fig. 14, pp. 521–523; §7, Fig. 9, pp. 513–514 (mark-up shocks; taxes in $1+\mu$); §4.4, p. 509.

Grading rubric.

- AS shifts up/left, y_e does not move, AD shift from the news only: 0.15 pt.
- Why AS is the constraint; substitution into L: 0.15 pt.
- Targeting rule $(y - y_e) + \phi(p - p^e) = 0$ from $dL/dy = 0$: 0.2 pt.
- y^*, p^*: 0.15 pt. i^* with its two components: 0.15 pt.
- Role of ϕ: news part independent of ϕ; mark-up part signed by $\phi - \sigma$; a limit case: 0.1 pt.
- Graph (AS_1, IT, AD before/after, O labelled): 0.1 pt.
- Differentiating in i with AD as the constraint, or targeting y_n instead of y_e: max. 0.4 pt. Claiming the whole response (news included) depends on ϕ versus σ: max. 0.6 pt.

Gap targeted: Prova 01, 3.c: $dL/dy = 0$ along AS not used; the response to news wrongly made to depend on ϕ.